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How to place the EEG-system correctly on the head

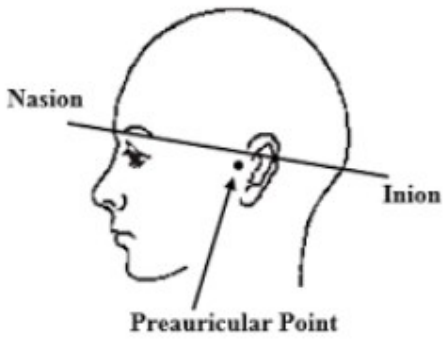
This article explains how to place the electrodes to be in agreement with the standard 10-10 system for electrode placement in EEG.

What is the ‘standard 10-10 system’?

The ‘standard 10-10 system’ is the academic standard for electrode placement in EEG, used by consensus in the entire world. (Almost) all EEG-systems available on the market follow this system – also EEG-systems which only use a few electrodes. For instance, the ABM B-Alert X10 system is an EEG-system with only 9 EEG-channels whereas the Brainproducts ActiCHamp is an EEG-system with 32 EEG-channels, but in both EEG-systems, the electrode location ‘Fz’ corresponds to exactly the same position on the scalp.

The system explained

For a precise placement of the electrodes, the ‘standard 10-10 system’ considers two distances on the head: the first one is the distance between the Nasion (i.e., the bony structure between the eyes) and the Inion (i.e., the bony structure in the back of the head) and the second one is the distance between the two ears (as measured from the preauricular point).



Both distances are then divided into 10 equal parts (hence the name ’10-10 system’), and the electrodes are placed on the intersections of these distances.

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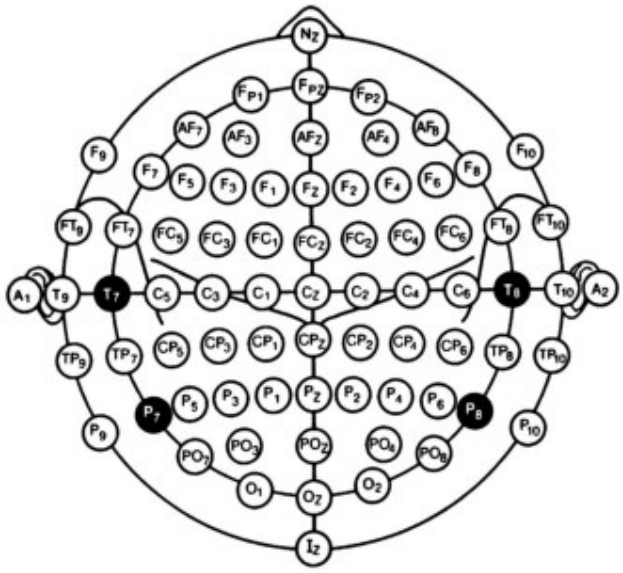


FIG. 1. Modified combinatorial nomenclature for the 10-10 system.

The electrode locations are further labeled in a standard way, as shown in the figure. The distances on the longitudinal axis are labeled by 'letters': Fp, AF, F, etc. (roughly corresponding to the underlying brain structures, for instance, with 'F' referring to Frontal cortex). The addition 'z' further indicates that the electrodes are located on the midline (dividing the two hemispheres), whereas even numbers are used for electrode locations in the right hemisphere, and uneven numbers for electrode locations in the left hemisphere.

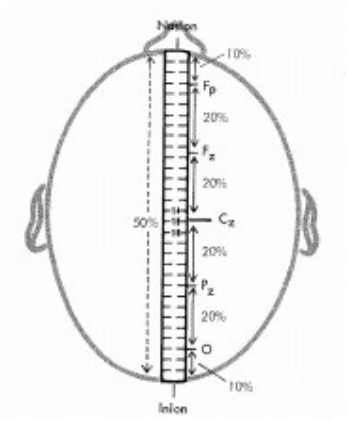
Click here for a science paper explaining the standard 10-10 system in further detail.

How to place the cap correctly on the head?

Obtaining a ‘perfect’ placement of the electrodes, that is, perfectly in accordance with the ‘standard 10-10 system’, is usually not feasible. This is because most EEG-systems are only available in a few standard sizes (e.g., S, M and L), or even in just one size – while the head sizes of individual respondents vary quite a lot. However, it is possible to approximate the best-possible placement by measuring the head sizes of the respondents.

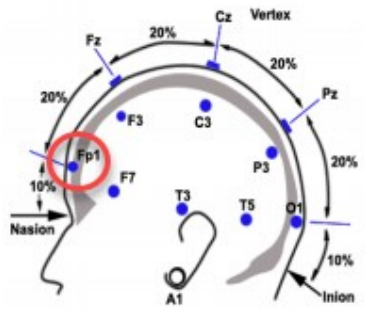
1. First measure the distance between the Nasion and Inion, and then, the distance between the two ears (from the preauricular point).

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2. Once these distances are known, choose the EEG-cap that is the best fit to the respondent's head size. (Verify the documentation provided by the hardware manufacturer for this).

3. Hereafter, choose a reference electrode for the placement of the head cap. Ideally, this is a frontal electrode on the midline axis – for instance: Fpz, AFz or Fz. Then, divide the distance between the nasion and inion by 10. Specifically, if the measured distance is 35 cm, dividing this distance by 10 gives you 3.5 cm. Fpz should be placed 3.5 cm (1 x 3.5 cm) away from the nasion, AFz should be placed 7 cm (i.e., 2 x 3.5 cm) away from the nasion, and Fz should be placed 10.5 (i.e., 3 x 3.5 cm) away from the nasion.



Hold the electrode at this location with one hand, and use the other hand to slide the cap over the head.

4. Finally, you choose another reference electrode. Ideally, this is a central electrode on the midline – for instance: Cz. Verify whether the distance between Cz and the left and right ears, respectively, is the same.

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Now, the EEG-system should be placed in accordance with the 'standard 10-10 system'.

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